2025年3月6日 9:03

1. $z=f(x+y,\ xy)$

$z=f(u,v)$ $\quad u=x+y,\quad v=xy$

$\frac{\partial z}{\partial x}=f_1'+f_2'y \qquad \frac{\partial z}{\partial y}=f_1'+f_2'\cdot x$

2. $z=f(x,u,v)$. $\quad u=g(x,y)$. $\quad v=h(x,y,u)$ 可微.

$\frac{\partial z}{\partial x}=f_x'+f_u'\cdot g_x'+f_v'\left(h_x'+h_y'\cdot 0+h_u'\cdot g_x'\right)$

$z=f(xy) \qquad z=f(t). \qquad t=xy$

$\frac{\partial z}{\partial x}=f'(xy)\cdot y$

第6讲. 向量值函数微分. 隐函数微分.

一.

设 $f:\mathbb{R}^n\to\mathbb{R}^m$, 若 $\forall\ X,Y\in\mathbb{R}^n$. $\forall\ \alpha,\beta\in\mathbb{R}$.

$$f(\alpha X+\beta Y)=\alpha f(X)+\beta f(Y)$$

则 称 f 线性映射.

设 $A_{m\times n}$ 矩阵. $\quad\forall\ x\in\mathbb{R}^n$.

$$f(x)=Ax. \qquad f:\mathbb{R}^n\to\mathbb{R}^m \text{ 线性映射.}$$

设 $f:\mathbb{R}^n\to\mathbb{R}^m$ 线性映射.

$e_1,\ e_2,\ \cdots,\ e_n\in\mathbb{R}^n. \qquad f(e_i)\in\mathbb{R}^m.$

$\forall\ x=\sum_{i=1}^n x_ie_i\in\mathbb{R}^n$.

$$f(x)=f\left(\sum_{i=1}^n x_ie_i\right)=\sum_{i=1}^n x_if(e_i)=\underbrace{\begin{pmatrix}f(e_1) & f(e_2) & \cdots & f(e_n)\end{pmatrix}}_{\parallel\ A_{m\times n}}\begin{pmatrix}x_1\\x_2\\\vdots\\x_n\end{pmatrix}$$

$$=Ax.$$

定理1. $\mathbb{R}^n\to\mathbb{R}^m$ 线性映射 与 $m\times n$ 阶 矩阵 一一对应.

$A_{m\times n}$.

$$\|A\|=\max\left\{\|Ax\|\ \middle|\ \|x\|=1,\ x\in\mathbb{R}^n\right\}<+\infty.$$

定理2. 线性映射$f:\mathbb{R}^n\to\mathbb{R}^m$ 是连续.

$\exists\ A_{m\times n}$.

$$f(x)=Ax.$$

$$\|f(x)-f(x_0)\|=\underbrace{\|A(x-x_0)\|\le\|A\|\cdot\|x-x_0\|}_{(x\to x_0)}\to 0.$$

$\forall\ x\in\mathbb{R}^n$.

$$\|Ax\|=\left\|A\frac{x}{\|x\|}\right\|\cdot\|x\|\le\|A\|\cdot\|x\|.$$

定义1. 设 $x_0=(x_{01},x_{02},\cdots,x_{0n})\in\mathbb{R}^n$. $\qquad\Delta x=(\Delta x_1,\Delta x_2,\cdots,\Delta x_n)$

$$\|AX\| = \|A \tfrac{X}{\|X\|}\| \cdot \|X\| \le \|A\| \cdot \|X\| .$$

定义1. 设 $X_0 = (x_{01}, x_{02}, \cdots, x_{0n}) \in \mathbb{R}^n$. $\Delta X = (\Delta x_1, \Delta x_2, \cdots, \Delta x_n)$.

若 $f: \mathbb{R}^n \to \mathbb{R}^m$ 在 X_0 全改变量

$$\Delta f = f(X_0 + \Delta X) - f(X_0) = \underline{A} \Delta X + \alpha(\Delta X)$$

其中 $A_{m\times n}$ 矩阵 与 ΔX 无关. $\lim\limits_{\|\Delta X\| \to 0} \dfrac{\|\alpha(\Delta X)\|}{\|\Delta X\|} = 0$.

则称 f 在 X_0 可微.

$\forall X = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$.

$$y = f(x_1, \cdots, x_n) \in \mathbb{R}^m$$

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix} = \begin{pmatrix} f_1(x_1, \cdots, x_n) \\ f_2(x_1, \cdots, x_n) \\ \vdots \\ f_m(x_1, \cdots, x_n) \end{pmatrix}$$

$$\begin{pmatrix} \Delta f_1 \\ \Delta f_2 \\ \vdots \\ \Delta f_m \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} \Delta x_1 \\ \Delta x_2 \\ \vdots \\ \Delta x_n \end{pmatrix} + \begin{pmatrix} \alpha_1(\Delta X) \\ \alpha_2(\Delta X) \\ \vdots \\ \alpha_m(\Delta X) \end{pmatrix}$$

$$\Delta f_1 = a_{11}\Delta x_1 + a_{12}\Delta x_2 + \cdots + a_{1n}\Delta x_n + \alpha_1(\Delta X)$$

$$\lim_{\|\Delta X\| \to 0} \frac{\|\alpha(\Delta X)\|}{\|\Delta X\|} = 0. \Rightarrow \lim_{\|\Delta X\| \to 0} \frac{|\alpha_1(\Delta X)|}{\|\Delta X\|} = 0.$$

f_i 在 X_0 可微 $\Longleftrightarrow$ f 在 X_0 可微.
$(i=1,2,\ldots m)$

$$a_{1j} = \frac{\partial f_1}{\partial x_j}\Big|_{X_0}$$

$$A = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & & & \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}\Bigg|_{X_0} = \frac{\partial(f_1, f_2, \cdots, f_m)}{\partial(x_1, x_2, \cdots, x_n)}\Bigg|_{X_0}$$

称为 f 在 X_0 Jacobi 矩阵. $(Jf(X_0)$ 或 $Df(X_0))$

$m \times n$.

$\therefore$ 称线性映射 $\Phi: \mathbb{R}^n \to \mathbb{R}^m$. $\Phi(X) = Df(X_0)X$

为 f 在 X_0 微分映射.

f 在 X_0 微分. $df(X_0) = Df(X_0)\begin{pmatrix} dx_1 \\ dx_2 \\ \vdots \\ dx_n \end{pmatrix}$

例2. $I \subset \mathbb{R}$ 区间. $x(t)$, $y(t)$, $z(t)$ 在 I 上可导.

空间曲线 L: $\begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases}$ $(\forall\ t \in I)$

可微 $f: I \to \mathbb{R}^3$. $\quad df(t_0) = \begin{pmatrix} x'(t_0) \\ y'(t_0) \\ z'(t_0) \end{pmatrix} dt \quad t_0 \in I$.

$$\begin{pmatrix} x'(t_0) \\ y'(t_0) \\ z'(t_0) \end{pmatrix} = \left.\frac{\partial(x,y,z)}{\partial t}\right|_{t_0}$$

例3. $D \subset \mathbb{R}^2$. 区域. $x(u,v)$, $y(u,v)$, $z(u,v) \in C^1(D)$.

曲面 S: $\begin{cases} x = x(u,v) \\ y = y(u,v) \\ z = z(u,v) \end{cases}$ $(u,v) \in D$

$f: D \to \mathbb{R}^3$. 可微. $\quad df(u_0, v_0) = \left.\frac{\partial(x,y,z)}{\partial(u,v)}\right|_{(u_0,v_0)} \begin{pmatrix} du \\ dv \end{pmatrix}$

二. 复合映射微分.

定理2. 设 $\Omega \subset \mathbb{R}^n$. $D \subset \mathbb{R}^m$ 非空.

设 $u = f(x): \Omega \to \mathbb{R}^m$. $\quad y = g(u): D \to \mathbb{R}^k$ 映射. 满足 $R(f) \subset D(g)$

若 f 在 $x_0 \in \Omega$ 可微. $\quad g$ 在 $u_0 = f(x_0)$ 可微.

则 复合映射 $g \circ f$ 在 x_0 可微. 且 $D(g \circ f)(x_0) = Dg(u_0) \cdot Df(x_0)$.

证: x_0. $\quad \Delta x$

$u_0 = f(x_0)$ $\quad \Delta u = f(x_0 + \Delta x) - f(x_0) = B \Delta x + \beta(\Delta x)$ $\quad \displaystyle\lim_{\|\Delta x\| \to 0} \frac{\|\beta(\Delta x)\|}{\|\Delta x\|} = 0$.

$y = g(u_0)$ $\quad \Delta y = g(u_0 + \Delta u) - g(u_0)$ $\quad B = Df(x_0)$

$m \times n$.

$$\begin{aligned} \Delta y &= (g \circ f)(x_0 + \Delta x) - (g \circ f)(x_0) \\ &= g(f(x_0 + \Delta x)) - g(u_0) \\ &= g(u_0 + \Delta u) - g(u_0) \\ &= A \Delta u + \alpha(\Delta u) \\ &= AB \Delta x + A\beta(\Delta x) + \alpha(\Delta u) \end{aligned}$$

$A = Dg(u_0)$ $\quad k \times m$. $\quad \displaystyle\lim_{\|\Delta u\| \to 0} \frac{\|\alpha(\Delta u)\|}{\|\Delta u\|} = 0$. $\quad \frac{\|\alpha(\Delta u)\|}{\|\Delta u\|} = \delta$

$\|\alpha(\Delta u)\| = \delta \|\Delta u\|$

$\displaystyle\lim_{\|\Delta u\| \to 0} \delta = 0$.

AB. $\quad k \times n$.

下证 $\displaystyle\lim_{\|\Delta x\| \to 0} \frac{\|A\beta(\Delta x) + \alpha(\Delta u)\|}{\|\Delta x\|} = 0$

$$0 \le \frac{\|A\beta(\Delta x) + \alpha(\Delta u)\|}{\|\Delta x\|} \le \|A\| \underbrace{\frac{\|\beta(\Delta x)\|}{\|\Delta x\|}}_{\to 0} + \frac{\|\alpha(\Delta u)\|}{\|\Delta x\|}$$

$\|\Delta x\| \to 0$

下证 $\displaystyle\lim_{\|\Delta x\| \to 0} \frac{\|\alpha(\Delta u)\|}{\|\Delta x\|} = 0$.

$$0 \le \frac{\|\alpha(\Delta u)\|}{\|\Delta x\|} = \delta \frac{\|\Delta u\|}{\|\Delta x\|} = \delta\left(\frac{\|B\Delta x + \beta(\Delta x)\|}{\|\Delta x\|}\right) \le \left(\|B\| + \frac{\|\beta(\Delta x)\|}{\|\Delta x\|}\right)\delta$$

$\|\Delta x\| \to 0 \Longrightarrow \|\Delta u\| \to 0$.

$$D(g\circ f)(x_0) = Dg(u_0)\cdot Df(x_0).$$

例4.
$$\begin{cases} y_1 = u_1u_2 - u_1u_3 \\ y_2 = u_1u_3 - u_2^2 \end{cases} \qquad \begin{cases} u_1 = x_1\cos x_2 + (x_1+x_2)^2 \\ u_2 = x_1\sin x_2 + x_1x_2 \\ u_3 = x_1^2 - x_1x_2 + x_2^2 \end{cases}$$

求 $\left.\dfrac{\partial(y_1, y_2)}{\partial(x_1, x_2)}\right|_{(1,0)}$

$f: (u_1, u_2, u_3) \longrightarrow (y_1, y_2)$ $\qquad$ $g: (x_1, x_2) \longrightarrow (u_1, u_2, u_3)$

$f\circ g: (x_1, x_2) \longrightarrow (y_1, y_2)$

$$\left.\frac{\partial(y_1, y_2)}{\partial(x_1, x_2)}\right|_{(1,0)} = \left.\frac{\partial(y_1, y_2)}{\partial(u_1, u_2, u_3)}\right|_{(2,0,1)} \cdot \left.\frac{\partial(u_1, u_2, u_3)}{\partial(x_1, x_2)}\right|_{(1,0)}$$

$(x_1, x_2) = (1, 0)$

$(u_1, u_2, u_3) = (2, 0, 1)$

三. 隐函数微分.

$F(x, y) = 0$. $\qquad$ $y = \varphi(x)$.

$F(x, \varphi(x)) = 0$.

$F_x' + F_y' \varphi'(x) = 0. \quad \Rightarrow \varphi'(x) = -\dfrac{F_x'}{F_y'}$

$F(x,y)=0$. $\quad (x,y) \in I \times J$. $\quad$ I. J 区间.

$\forall x \in I$, $\exists! y \in J$. s.t. $F(x,y)=0$. 则由 $F(x,y)=0$ 确定隐函数 $y=\varphi(x)$.

$$\begin{cases} z = F(x,y) \\ z = 0 \end{cases}$$

定理3. 设 $F(x,y)$ 定义在开集 $D \subset \mathbb{R}^2$ 上, 满足

① $F(x,y) \in C^1(D)$.

② $\exists (x_0,y_0) \in D$. s.t. $F(x_0,y_0)=0$ $\qquad F(x,y)=0$.

③ $\left.\dfrac{\partial F}{\partial y}\right|_{(x_0,y_0)} \neq 0$. $\quad \left(\left.\dfrac{\partial F}{\partial x}\right|_{(x_0,y_0)} \neq 0 \right)$

则 $\exists I \times J \subset D$. s.t. $(x_0,y_0) \in I \times J$. $\quad$ (I. J 开区间). 满足

① $\forall x \in I$, $\exists! y \in J$ s.t. $F(x,y)=0$. 即由 $F(x,y)=0$ 确定隐函数 $y=\varphi(x)$.

② $\varphi(x_0) = y_0$.

③ $y = \varphi(x) \in C^1(I)$. 且 $\varphi'(x) = -\dfrac{F_x'(x,y)}{F_y'(x,y)}$.

证明: 设 $\left.\dfrac{\partial F}{\partial y}\right|_{(x_0,y_0)} > 0$, $\qquad F(x,y) \in C^1(D)$.

$\therefore \exists I_1 . J_1$. $\quad (x_0,y_0) \in I_1 \times J_1$. s.t. $\forall (x,y) \in I_1 \times J_1$.

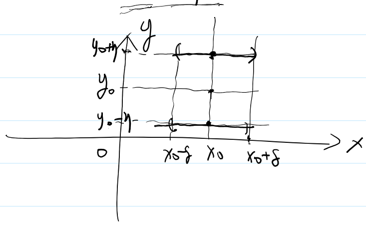
$\dfrac{\partial F}{\partial y}(x,y) > 0$. $\qquad I_1 = (x_0-\delta_1,\ x_0+\delta_1)$

$J_1 = (y_0-\eta_1,\ y_0+\eta_1)$

$\therefore F(x,y)$ 关于 y 在 $I_1 \times J_1$ 上严单增.

$\because F(x_0,y_0)=0$.

$\therefore \forall \eta < \eta_1$. $\quad \therefore F(x_0, y_0+\eta) > 0$, $\quad F(x_0, y_0-\eta) < 0$



$\because F(x,y) \in C^1(D)$

$\because$ $F(x,y)$ 关于 x、y 分别连续.

$\exists\, \delta>0$. $(\delta<\delta_1)$. s.t. $\forall x\in(x_0-\delta,\ x_0+\delta)$ $F(x,\ y_0+\eta)>0$. 且 $F(x,\ y_0-\eta)<0$.

$\exists!\, y\in(y_0-\eta,\ y_0+\eta)$ s.t. $F(x,y)=0$.

$y=\varphi(x)$

$\forall \eta>0$. $\exists\,\delta>0$. s.t. $\forall x\in(x_0-\delta.\ x_0+\delta)$. 有 $|y-y_0|<\eta$.

$$|y-y_0| = |\varphi(x)-\varphi(x_0)|$$

$\varphi(x)$ 在 x_0 连续.

②. 下证 $y=\varphi(x)\in C^1(I)$.

下证 $y=\varphi(x)\in C(I)$. $I=(x_0-\delta,\ x_0+\delta)$

即证 $\forall x_1\in I$. $y=\varphi(x)$ 在 x_1 连续. $y_1=\varphi(x_1)$

$(x_1,y_1)\in D$.

$F(x_1,y_1)=0$. $\left.\dfrac{\partial F}{\partial y}\right|_{(x_1,y_1)}>0$.

$\therefore$ $y=f(x)$ $I_2=(x_1-\delta_2,\ x_1+\delta_2)\subset I$ $J_2=(y_1-\eta_2,\ y_1+\eta_2)\subset J$

$f(x)$ 在 x_1 连续.

$\forall x\in I_2$. $f(x)=\varphi(x)$ 在 x_1 连续.

$\because$ $F(x,y)\in C^1(D)$. $\forall (x,y)\in D$.

$$\Delta F = F_x'(x,y)\Delta x + F_y'(x,y)\Delta y + \varepsilon_1\Delta x+\varepsilon_2\Delta y \qquad \lim_{(\Delta x,\Delta y)\to(0,0)}\varepsilon_i=0.$$

$y=\varphi(x)$. $F(x,\varphi(x))\equiv 0$.

$\Delta F=0$.

$$\frac{\Delta y}{\Delta x} = -\frac{F_x'(x,y)+\varepsilon_1}{F_y'(x,y)+\varepsilon_2} \longrightarrow -\frac{F_x'(x,y)}{F_y'(x,y)}$$

$\Delta x\to 0$. $\Rightarrow$ $\Delta y\to 0$.

$$\therefore\ \varphi'(x) = -\frac{F_x'(x,y)}{F_y'(x,y)}$$

$F(x,y) \in C^k(D)$. $\Rightarrow$ $y=\varphi(x) \in C^k(I)$.

例1. $\sin x + \ln y - xy^3 = 0$. 在 $(0,1)$ 附近是否确定隐函数.

解: $F(x,y) = \sin x + \ln y - xy^3$.

$F_x', F_y' \in C^k$.

$F_x'(0,1) = \cos x - y^3 \big|_{(0,1)} = 0$.

$F_y'(0,1) = \frac{1}{y} - 3xy^2 \big|_{(0,1)} = 1$

$\therefore$ $F(x,y)=0$. $y = f(x) \in C^k$. (显式)

推论. 若 $\underbrace{F(x,y)}_{n+1}$ $(x \in \mathbb{R}^n, y \in \mathbb{R})$ 在 $(x_0, y_0) \in \mathbb{R}^{n+1}$ 邻域 W 内有定义.

① $F \in C^k(W)$. $(k \geq 1)$

② $F(x_0, y_0) = 0$.

③ $\frac{\partial F}{\partial y}\Big|_{(x_0, y_0)} \neq 0$.

则 $\exists \delta > 0$, $\eta > 0$. s.t. $B(x_0, \delta) \times (y_0 - \eta, y_0 + \eta) \subset W$.

且 $\forall x \in B(x_0, \delta)$. $\exists ! y \in (y_0 - \eta, y_0 + \eta)$, s.t. $F(x, y) = 0$.

$\underline{y = f(x)}$ 满足

① $y_0 = f(x_0)$

② $y = f(x) \in C^k(B(x_0, \delta))$ 且 $\frac{\partial y}{\partial x_i}\Big|_x = -\frac{F_{x_i}'(x,y)}{F_y'(x,y)}$

$\forall x \in B(x_0, \delta)$.

$F(x_1, x_2, \cdots, x_n, \overset{y}{f(x_1, \cdots, x_n)}) \equiv 0$.

$F_{x_i}' + F_y' \cdot f_{x_i}' = 0$.